MIL-L-45589

2 February 1970

SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

LAUNCHER, GRENADE, 40MM; M79

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of single-shot, breech-loading, shoulder weapon which fires a low velocity 40 millimeter (MM) grenade projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-	Preservation, Methods of.
MIL-A-8625	-	Anodic CoatingS for Aluminum and Aluminum Alloys.
MIL-G-10944	-	Gages, Dimensional Control.
MIL-W-13855	-	Weapons Small Arms and Aircraft Armament Subsystem, General Specification for.
MIL-P-16232	-	Phosphate Coatings, Heavy, Manganese of Zinc Base (For Ferrous Metals).
MIL-L-45607	-	Inspection Equipment, Supply and Maintainance of.
MIL-C-45662	-	Calibration System Requirements.

STANDARDS

Military

MIL-STD-9	-	Screw Thread Conventions and Methods of Specifying.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1010

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DRAWINGS

U.S. Army Weapons Command
F7790600 - Launcher, Grenade, M79.
IEL 7274616- List of Inspection Equipment Numbers.

(Part drawings referenced in this specification form apart of Drawing F7790600; inspection equipment drawings forma part of Drawing IEL 7274615.)

PUBLICATIONS

U.S. Army Weapons Command
P8426167 - Packaging Data Sheet for Launcher, Grenade, 40MM, M79 W/Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute Incorporated
B46.1 - Surface Texture
Y14.5 - Dimensioning and Tolerancing for Engineering Drawings

(Application for copies should be addressed to the American National Standards Institute Incorporated, 1430 Broadway, New York, New York 10018.)

American Welding Society
Standard AWS A2.0 - Welding Symbols

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Quality evaluation items. On each contract samples of launchers, parts and pilot pack representing initial production shall be forwarded to the designated testing agency (see 6.1) for quality evaluation. Launchers and packaging shall have been inspected and determined acceptable.

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3.2 Materials. Launchers and parts shall conform to the material requirements specified herein, on drawings referenced on Drawing F7790600 and shall be in accordance with the applicable material provisions of MIL-W-13855.

3.3 Design. Launchers and parts shall conform to the design requirements specified herein and on drawings referenced on Drawing F7790600 and shall be in accordance with the applicable design provisions of MIL-W-13855.

3.3.1 Barrels. Barrels shall be made from aluminum alloy material as specified on Drawing F7791569. Each barrel shall be proof tested by firing one standard cartridge. There shall be no visible evidence of failure. Parts which satisfactorily meet this requirement shall be marked by stamping the letter "P" as indicated on the drawing.

3.3.2 Barrel with fore end assembly. The barrel, with fore end assembly attached, shall pivot on the fulcrum pin through its full range of travel without binding when the launcher is opened and closed.

3.3.3 Extractor. The extractor shall be retained in the extractor hole in the barrel by the spring pin and shall function through its full range of travel under spring action without binding. When the launcher is opened, the extractor shall move to the snapped-out position and push the cartridge case into position to permit removal from the chamber. The extractor shall return to its seated position without rough movement, when in contact with the breech face and firing pin retainer, when the launcher is closed.

3.3.4 Stock and fore end. When the stock screw is drawn up with the torque specified on the applicable drawing, the arms of the stock shall be in positive contact with the mating recoil seat of the receiver so that there shall be no movement of the stock relative to the tang of the receiver. The fore end shall be fastened securely by screws to the fore end bracket and sight base so that there shall be no relative movement.

3.3.4.1 Plastic stick. Stocks shall be free of cracks and holes. Joints shall be free of unbended areas, voids, or excess adhesive and shall be such that adjacent contours of the two body halves of the stock shall be uniformly blended. The surface of the two screw holes filled with the resin mixture shall be level or recessed within .015 inch of the outside surface. The plastic stocks shall be in accordance with Drawing F11010344.

3.3.5 Sling swivel assemblies. The front swivel assembly shall be fastened securely in the mating cutout of the fore end by screws to the fore end bracket and sight base. The rear swivel assembly shall be fastened securely in the mating cutout of the stock by the screws. Screws shall be torqued to meet the requirements of the respective assembly drawings. There shall be no relative movement of these parts except that the swivel loops shall rotate through their full range of travel with no lateral movement beyond the limits specified on the applicable drawing.

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3.3.6 Recoil pad. The recoil pad shall be fastened securely to the stock by screws. Screws shall be torqued as specified on Drawing D11010343 after alining to the stock.

3.3.7 Latch. The latch shall be fastened securely to the latch pivot by the spring pin and shall move through its full range of travel without binding. The latch shall pivot under thumb pressure to the "open" position and shall be held in the "open" position by spring action of the latch lock, permitting the launcher to be opened. The latch shall remain in the "open" position until the launcher is closed. Closing the launcher shall cause the locking lug to depress the latch lock. The latch shall return under spring action to its original position, locking the barrel in battery position. It shall not be possible to open the launcher until the latch has been moved to the "open" position. The latch lock shall travel freely without bind. The latch shall operate with the launcher held in any position.

3.4 Heat treatment. When heat treatment is specified it shall be in accordance with that specified on the applicable drawing.

3.5 Dimensions and tolerances. Dimensions and tolerances of parts and assemblies of the launcher shall conform to the detail drawings referenced on Drawing F7790600. Interpretation of dimensions and tolerances shall be in accordance with Y14.5. For phosphate coated parts, the dimensional provisions of MIL-P-16232 shall apply and for anodized coated parts, the dimensional provisions of MIL-A-8625 shall apply.

3.6 Surface roughness. Surface roughness shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be as defined in B46.1.

3.7 Protective finishes.

3.7.1 Inorganic finishes. Parts requiring an inorganic finish shall be finished or coated as indicated on the applicable drawing and in accordance with MIL-STD-171.

3.7.2 Painting. Launcher parts shall be painted as specified on the applicable drawing. Touch-up procedures shall be in accordance with MIL-W-13855. Parts not to be painted shall be in accordance with MIL-STD-171.

3.8 Construction.

3.8.1 Assembly. Parts and components shall be positioned and attached as depicted on Assembly Drawing F7790600 and applicable detail drawings.

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3.8.2 Welding and brazing. Welding and brazing shall be in accordante with the applicable drawing. Interpretation of welding symbols depitted on the applicable drawings shall be as defined in AWS A2.O.

3.8.3 Threads. Threads shall conform to the thread series and class indicated on the applicable drawing and in accordance with MIL-STD-9.

3.8.4 Identification. Marking of launchers, including application of part numbers, shall be as indicated on the applicable drawing and in accordance with MIL-W-13855. Serial numbers shall be furnished by the procuring activity.

3.9 Performance and product characteristics.

3.9.1 Safety.

3.9.1.1 Functioning of safety. When the barrel is locked in safety position and the trigger is in the normal forward position, the safety shall be movable between the "safe" and "fire" positions and shall remain in the position set under spring action of the safety spring plunger until reset manually. When the safety is positioned forward, so that the letter "F" is completely uncovered and the hammer is cocked, the hammer shall release as the trigger is pulled. When the safety is positioned rearward, so that the letter "S" is completely uncovered, the safety bar shall restrict trigger movement and prevent firing of the launcher. When the safety is in "fire" position, rotating the latch to any point where the latch permits the launcher to open shall cause the safety to move to the "safe" position.

3.9.1.2 Safety actuation force. The force required to move the safety from the safe position to the "fire" position shall be not greater than 16 pounds, nor less than 4 pounds, when tested as specified in 4.4.3.1 and 4.5.7.

3.9.1.3 Sear safety. The difference between the maximum vertical movement of the sear during cocking of the hammer and the maximum vertical movement of the sear, when the trigger is pulled with the hammer cocked and the safety in the "safe" position, shall be not less than 0.026 inches at datum -A- shown on Drawing F7790600, when tested as specified in 4.4.3.1 and 4.5.8.

3.9.2 Hammer. The hammer shall pivot on the hammer pin through its full range of travel without binding. When the launcher is fully opened, the hammer shall move to the cocked position by action of the cocking arm on the cocking lever and shall be held in the cocked position by engagement with the sear. When the safety is set in "fire" position, the hammer shall disengage from the sear as the trigger is pulled and move under spring action to actuate the firing pin. After actuating the firing pin, the hammer shall be returned to, and be held in, its normal forward position by spring action on the sear, permitting the firing pin nose to return to its retracted Position, flush with, or below, the breech face.

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3.9.3 Trigger and sear. With the hammer cocked, and the safety in "fire" position, the trigger and sear shall return to their original positions under spring action after any partial trigger pull short of releasing the hammer. The trigger and sear shall also return to their Original positions after any trigger pull with the hammer in the cocked or released position.

3.9.3.1 Trigger guard assembly. The trigger guard assembly shall be retained on the receiver by the staked trigger guard bracket screw and by assembly with the detent assembly, detent spring, and detent spring housing. When the detent spring housing is fully depressed, the trigger guard shall be capable of being manually rotated on the trigger guard rivet and the detent plunger between the normal vertical position of the guard and both open positions, with only slight bind. The spring actuated detent assembly shall lock the guard in any of the three positions when pressure on the detent spring housing is released and the locking hole is rotated into alinement with the detent.

3.9.4 Firing pin. The firing pin shall be retained in the receiver by the firing pin retainer, which shall be drawn up to the torque specified on the drawing. The firing pin shall move though its full range of travel under spring action without binding. When the firing pin is moved forward through partial or full travel, it shall be returned to, and held in, its original retracted position by spring tension so that the point is flush with, or below, the breech face.

3.9.5 Sight assembly.

3.9.5.1 The sight base shall be oriented with respect to the barrel and sight base index marks and shall fit snugly on the barrel so that there shall be no relative movement after the sight base screw is drawn up with the torque specified on the applicable drawing and the head of the screw is drawn up with the torque specified on the applicable drawing and the head of the screw is positioned in the radical groove on the barrel.

3.9.5.2 When the sight lock is depressed, the frame base and frame assembly with attached components shall be capable of being manually rotated, without bind, on the windage screw between the upright locking position and the forward or folded-down locking position. The frame base shall lock into each position under spring action of the sight lock when pressure on the lock is released.

3.9.5.3 When the retainer lock nut is tightened, it shall lock the aperture carrier, together with the sight aperture and aperture retainer, in any position on the frame assembly. When the retainer lock nut is loosened and depressed, the carrier, aperture, and retainer shall be capable of being moved manually on the frame assembly through their full range of adjustment. When the retainer locknut is released, the carrier, aperture, and retainer shall be held in position under spring tension. When the

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retainer locknut is loosened or tightened, the elevating wheel for micro-elevation adjustment of the sight aperture shall be capable of being manually rotated causing the carrier, aperture, and retainer to move on the frame assembly through the full range of adjustment. The elevating wheel shall rotate freely between indexing positions and shall be restrained in the indexing positions by the spring actuated indexing plunger. When the frame assembly is in the upright position, and the carrier, aperture, and retainer are moved vertically the sight aperture shall maintain contact with the sight scale under spring tension and shall move laterally on the aperture retaining screws.

3.9.5.4 After the setscrews on the frame assembly have been adjusted and staked as specified on the applicable drawing the windage screw shall be capable of being rotated manually causing the frame assembly, with attachment components, to move laterally through the fill range of adjustment. The windage screw shall rotate freely between indexing positions and shall be restrained in the indexing positions by the spring actuated indexing plunger.

3.9.5.5 When the scale lock screw is tightened, the spring pin shall be positioned in the angular location specified on the applicable drawing and the sight scale shall be locked in position on the sight frame. When the screw is loosened, the scale shall slide through its full range of adjustment.

3.9.6 Front sight. The front sight shall be capable of lateral adjustment on the tenon of the barrel and, when adjusted, shall be locked securely in position by the front sight screw. There shall be no relative movement of these locked parts.

3.9.7 Receiver group. The hammer pin, safety bar pin, trigger pin, and safety spring shall be retained securely by staking the receiver as specified on the applicable drawings. With the stock assembly removed, it shall not be possible to dislodge the hammer pin with the hammer in both the cocked or uncocked positions, or safety bar pin by striking each side of the receiver sharply against the palm of the hand and it shall not be possible to unseat the safety spring by operating the safety.

3.9,8 Cocking lever and cocking arm. When the barrel of the launcher is opened fully, the amount of engagement between the cocking lever and the cocking arm, as depicted on Drawing F7790600, shall be not less than 3/16 of an inch.

3.9.9 Receiver and fulcrum pin. The straight headless (fulcrum) pin shall be assembled securely to the receiver so that there shall be no relative movement when an axial load of 200 pounds is applied gradually to the pin.

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3.9.10 Firing pin indent. The firing pin indent shall be 0.008 to 0.011 inches and shall not be off-center more than one-half the diameter of the firing pinpoint when tested as specified in 4.4.3.2 and 4.5.1.

3.9.11 Firing pin intrusion. The firing pin intrusion shall be 0.000 to 0.009 inches when tested as specified ln 4.4.3.2 and 4.5.2.

3.9.12 Trigger pull. The trigger pull shall be free of creep and shall be greater than 6.5 pounds but shall not exceed l.1 pounds when tested as specified in 4.4.3.1 and 4.5.3. Creep shall be interpreted to mean any perceptible rough movement between the time the trigger slack is taken up and the hammer is released.

3.9.13 Interchangeability. Unless otherwise indicated on the drawings, parts shall be interchangeable. (In normal assembly operations, there shall be no objections to preferential assembly of parts provided that all parts are dimensionally acceptable.) Launchers and repair parts shall be capable of meeting the interchangeability tests specified in 4.4.3.3 and 4.5.4.

3.9.14 Pressure resistance. Launchers shall be capable of withstanding the pressure resistance test (proof firing) specified in 4.4.3.1, 4.5.5 and 4.5.5.1. Parts shall be free of cracks, seams, and other defects after proof firing as evidenced by visual examination.

3.9.15 Functioning. Launchers shall operate without malfunctions and unserviceable parts when tested as specified in 4.4.3.1, 4.5.5 and 4.5.5.2.

3.9.16 Alinement of sights.

3.9.16.1 The sight scale shall-be capable of being adjusted so that the angle between the line of sight established by the 200 meter scale mark and a horizontal plane containing the center line of the barrel bore is 208 roils. With the sights set for the 200 meter range, the windage scale set at zero, and the center line of the barrel bore perpendicular to a target, the front sight shall be adjusted so that the intersections with the target of the projected center line of the barrel bore and the line of sight established by the 200 meter scale mark are horizontally displaced on the target by .44 inches per 100 inches of target distance, and locked. Target distance is defined as the distance from the forward tip of the front sight to the target, when measured parallel to the bore center line.

3.9.16.2 With the sight components adjusted as specified in 3.9.16.1, the angle between the line of sight established by the 75 meter scale mark and a horizontal plane containing the bore center line shall be 69 mils ± 3 mils. The angle between the line of sight established by the 350 meter scale mark and a horizontal plane containing the bore center line shall be 476 mils ± 9 mils.

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The intersections with the target of the line of sight, based on scale setting and the projected center line of the bore, shall be laterally displaced by zero inches ±.25 inches per hundred inches of target distance fir the 75 meter scale setting and by 1.43 inches ± .25 inches per hundred inches of target distance for the 350 meter scale setting, with target distance as defined in 3.9.16.1.

3.9.17 Endurance. Launchers shall be capable of passing a 1000 round endurance test without malfunctions or unserviceable parts when tested as specified in 4.4.3.4, 4.5.5 and 4.5.5.3.

3.10 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or Purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Quality evaluation item inspection. Inspection of quality evaluation launchers (see 3.1) and pilot pack (see 5.1) shall be performed at the designated testing agency. Launchers shall be subjected to the examination and test specified herein including function firing and endurance firing and such other examination and tests as are necessary to determine compliance with the requirement of the contract. Level A unit packages shall be subjected to the free fall drop test specified in MIL-P-116 except that drop height shall be 21 inches.

4.4 Quality conformance inspection.

4.4.1 Inspection lot.

4.4.1.1 Launchers. Unless otherwise specified, an inspection lot of launchers shall be one month's production. Launchers shall be assembled from lots of component parts that have met all inspection requirements. specified herein.

4.4.1.2 Parts and packaging. The formation, size, and presentation of inspection lots of parts and packaging shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions, contractor's delivery schedule, and within the limitations of MIL-W-13855.

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4.4.2 Examination.

4.4.2.1 Parts and packaging.

4.4.2.1.1 Component parts and concurrent repair parts. Examination of parts shall be performed as specified herein and in accordance with the criteria specified in the contract (see 6.1). The contractors examination of parts shall be accomplished prior to their assembly into the launcher or submission for acceptance as repair parts.

4.4.2.1.2 Packaging. Examination of packaging of launchers shall be performed in accordance with 4.4.2.1.2.1 (see 6.1). Examination for packaging defects specified in 4.4.2.1.2.1 shall apply to each item of the applicable sample of launchers, interior packages, or exterior containers, as applicable. Sample size shall be in accordance with MIL-STD 105 using inspection level I. Examination of packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

4.4.2.1.2.1 Packaging. Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P8426167.

Categories	Defect	AQL
Critical:	None defined.	
Major:		
101	Eligible or incorrect marking.	1.0
102	Missing parts or items of equipment	1.0
103	Improper level packaging or packing (see procurement documents).	1.0
104	Inadequate cleaning and drying.	1.5
105	Improper preservative application and drainage.	1.5
106	Improper cushioning and wrapping.	1.5
107	Improper closure of interior packages.	1.5
108	Improper bags or containers.	1.5
109	Improper positioning of launcher or equipment in interior package.	1.5
110	Improper closure and strapping of shipping containers.	1.5
Minor:	None defined.	

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4.4.2.2 Launchers. Final examination of launchers shall be performed after completion of all testing and just prior to preservation and packaging. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating. Launchers failing to meet the requirements shall be rejected. Each launcher shall be examined as specified below.

4.4.2.2.1 Visually examine launcher for cleanliness, workmanship, and completeness of manufacture assembly, finish and marking (identification and proof firing).Manually examine launcher for functioning of operating parts and visually examine markings for clarity and legibility. Visually and manually examine the following parts and assemblies to determine conformance to the applicable requirements:

Part or assembly	Reference paragraph
Barrel	3.3.1
Barrel with fore end assembly	3.3.2
Extractor	3.3.3
Stock and fore end	3.3.4
Sling swivel assemblies	3.3.5
Recoil pad	3.3.6
safety	3.9.1
Hammer	3.9.2
Trigger guard assembly	3.9.3.1
Firing pin	3.9.4
Sight assembly	3.9.5
Front sight	3.9.6
Cocking lever and cocking arm	3.9.8

4.4.2.2.2 Latch.

(a) Visually and manually examine latch for compliance with 3.3.7.
(b) Manually examine to assure that launcher does not open until latch has been moved to and locked in the “open” position by a gradual uninterrupted application of thumb pressure to the latch.

4.4.2.2.3 Trigger and sear.

(a) Disassemble stock from receiver.
(b) Visually and manually examine trigger and sear for compliance with 3.9.3.

4.4.2.2.4 Receiver group and fulcrum pin.

(a) Manually examine receiver group for compliance with 3.9.7.
(b) Manually examine receiver and falcrum pin (hawing A7790637) to determine compliance with 3.9.9 by applying the load specified on Drawing 7790600 with a force gages

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(c) Reassemble stock to receiver assuring that stock screw is drawn up with the required torque (see Drawing F7790600).

4.4.2.2.5 Launchers which have passed all examinations after successful completion of testing shall be stamped by the contractor with the Department of Defense complete inspection approval stamp as specified on the applicable drawing. The Government representative will control the stamps used for such stampings and observe the stamping operations as required to assure successful completion of all required examinations and tests prior to stamping.

4.4.3 Testing.

4.4.3.1 Triggerpull, pressure resistance, functioning, alinement of sights, safety actuation force, and sear safety testing. The contractor shall test each launcher for trigger pull, pressure resistance functioning, alinement of sights, safety actuation force, and sear safety using the test methods specified in 4.5.3, 4.5.5 (as applicable), 4.5.6, 4.5.7, and 4.5.8 respectively. Launchers failing to meet any of the requirements shall be rejected.

4.4.3.2 Firing pin indent and firing pin intrusion testing. The contractor shall test a sample of 10 1aunchers selected by the Government representative from each inspection lot for firing pin indent and firing pin intrusion using the test methods specified in 4.5.1 and 4.5.2 respectively. Failure of any launcher in the sample to meet the requirements shall cause rejection of the represented lot.

4.4.3.3 Interchangeability testing.

4.4.3.3.1 In plant.

4.4.3.3.1.1 Launchers. Ten launchers selected by the Government representative from inpection lot shall be tested by the contractor for interchangeability using the test methods specified in 4.5.4.1.1. Launchers taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. The 10 launchers shall be tested for and shall comply with the requirements for firing pin indent and firing pin protrusion before and after interchange of parts using the test methods specified in 4.5.1 and 4.5.2 respectively. In addition, the launchers shall be examined for functioning of safety (see 3.9.1.1) and shall be tested for functioning, alinement of sights, safety actuation force, and sear safety requirements after interchange of parts using the test methods specified in 4.5.5, 4.5.6, 4.5.7, and 4.5.8 as applicable. Hand refinement of parts will not be allowed on any launchers during interchange of parts, except as noted on assembly drawing. No malfunctions or failures shall be allowed in the functioning of safety examination or in the functioning, alinement of sights, safety actuation force, and sear safety tests. Failure of the interchangeability test shall cause retest or rejection of the represented lot. At the discretion of the

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Government representative, an interchangeability retest may be allowed without reconditioning the lot of launchers. Failure in the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of 20 launchers from each retest or reconditioned lot shall be tested using the same procedure described above.

4.4.3.3.1.2 Concurrent repair parts. The contractor shall subject at least two parts from each inspection lot of current repair parts to the interchangeability test specified in 4.5.4.1.2. Failure of any part to meet the requirements shall be cause for rejection of the represented lot of parts subject to reconditioning and further test as a reconditioned lot. A sample of double the number of parts used in the original test shall be tested from each reconditioned lot using the test method specified in 4.5.4.1.2.

4.4.3.3.2 Interplant. When launchers are manufactured concurrently by more than one contractor, each contractor shall forward monthly to a testing agency designated by the procuring agency, five launchers for the interplant interchangeability test specified in 4.5.4.2 (see 6.1). The contractor will be informed of any failure of the launchers to meet prescribed requirements. Upon completion of inspection by the testing agency, samples may be commercially packaged and will be returned to the contractor for reexamination and repackaging in accordance with procurement documents at the contractor's expense.

4.4.3.4 Endurance testing. One launcher selected by the Government representative from each months production shall be tested by the contractor for endurance using the test methods specified in 4,5.5 and 4.5.5.3. If the endurance requirements of 3.9.17 are not met, the represented lot shall be rejected and the contractor shall:

4.4.3.4.1 Conduct a failure analysis by perfoming a dimensional, physical and visual examination (as applicable of the parts which are suspected to be cause of the failed components or malfunctions.

4.4.3.4.2 Within two (2) weeks after completing the failure analysis study, forward a report of the study to the quality Assurance Representative and to Commanding General, U.S. Army Weapons Command, Attn: AMSWE-QA, Rock Island, Illinois 61201.

4.4.3.4.3 Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defects in future production.

4.4.3.4.4 Examine launchers, partially assembled launchers, and components (including parts and subassemblies at either or both in-process or final assembly) to insure that material containing the same or similar defects is purged from the inventory and not presented to the Government for acceptance.

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4.4.3.4.5 Upon instruction from AMSWE-QA, resume testing of the launcher which failed the endurance test, or retest an additional sample of two (2) launchers from same lot or reconditioned lot for endurance testing. In the event the cause of the failed components or malfunctions cannot be determined the contractor shall request through the Administrative Contracting Officer, technical assistance from the procuring activity.

4.4.3.5 Certification. The contractor shall be responsible for testing necessary to accomplish certification. For each inspection lot of launchers, the contractor shall provide the Government representative with certified statements of compliance with applicable drawings and specifications for the following materials and processes:

(a) Touch-up procedures (see MIL-W-13855).
(b) Torque of screws (see Drawing F7790600).
(c) Pin, Straight, Headless (see Drawing A7790637).

4.4.3.6 Component parts and concurrent repair parts testing. Raw material testing and part testing shall be performed in accordance with the criteria specified in the contract (see 6.1). Statements of finding will include chemical analysis and physical tests of materials, and tests of protective finish, heat treatment, bonding, and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item or submission for acceptance as repair parts.

4.4.3.7 Packaging testing.

4.4.3.7.1 Launchers.

4.4.3.7.1.1 The contractor shall furnish the Government representative with certification that the packaging materials conform to the applicable packaging data sheets and specifications.

4.4.3.7.1.2 Determination of cleanliness. The contractor shall test items from each inspection lot for cleanliness using the test methods specified in 4.5.9.1. Sampling shall be in accordance with MIL-P-116.

4.4.3.7.1.3 Heat-sealed seam and vacuum retention. The contractor shall test level A unit packages from each inspection lot for heat-sealed seam and vacuum retention using the test methods specified in 4.5.9.2 and 4.5.9.3 respectively. Sampling shall be in a ccordance with MIL-P-116.

4.4.3.7.2 Repair parts. Testing of packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

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4.4.4 Inspection equipment.

4.4.4.1 Unless otherwise specified in the contract (see 6.1), responsibilities for acquisition, maintenance, and disposition of inspection and test equipment prescribed on list of Inspection Equipment Numbers IEL 7274616, and for all other inspection equipment required to perform inspection prescribed by applicable specifications) shall be in accordance with MIL-I-45607 and MIL-C-45662.

4.4.4.2 Unless otherwise specified in procurement documents, Government standard 40MM M378 test inert projectile) cartridges shall. be used in all firing tests (see 6.1).

4.5 Test methods.

4.5.1 Firing pin indent test. Launchers shall be tested for firing pin indent requirement (see 3.9.10) using the inspection equipment in accordance with Drawings C7218984 and C7798912. The launcher shall be held in a horizontal position (approximately), the launcher opened, and the copper compression cylinder holding fixture containing the copper compression cylinder, Drawing B5039436, inserted in the barrel chamber. The launcher shall then be closed, the safety set in the “fire” position, and the trigger pulled to release the hammer and indent the copper cylinder. The holding fixture shall then be removed from the launcher and the depth of the indent in the copper compression cylinder computed by measuring the distance from the original surface of the copper cylinder (before indentation) to the bottom of the firing pin impression. If any firing pin indent is not within the requirements, three more impressions shall be taken and the average depth of the three indents of each test shall be within the requirements. Each indent shall be visually examined to determine whether the concentricity requirement has been met.

4.5.2 Firing pin intrusion test. Launchers shall be tested for firing pin intrusion requirement (see 3.9.11) using the inspection equipment in accordance with Drawing D7799026. The test shall be accomplished with the barrel removed and the hammer cocked.

4.5.3 Trigger pull test. Launchers shall be tested for trigger pull requirement (see 3.9.12) using inspection equipment in accordance with Drawing D7799050 and D7799051. The launcher shall be cocked, the safety shall be in the fire position, and the launcher shall be in the vertical position with the stock assembly down when each load is applied. The load shall be applied gradually to the trigger and exerted in a line parallel to the axis of the barrel bore by a slow steady lift of the launcher. When the minimum load is applied the hammer shall not release, and when the maximum load is applied the hammer shall release. The trigger pull shall also be tested for creep by applying pressure manually to the trigger at a uniform rate of increase over a period of not less than 3 seconds.

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4.5.4 Interchange of parts.

4.5.4.1 In plant.

4.5.4.1.1 Launchers. Launchers shall be tested for interchange of parts (see 3.9.13) by disassembling and then reassembling parts using the parts and prearranged system prescribed below. Interchange of parts shall be accomplished by divising the parts of each launcher into 10 groups of nonmating parts as shown below and distributing the groups into 10 different trays until each tray contains a complete launcher. Groups of parts from launcher number 1 shall be taken in order and placed in trays 1 through 10: groups of parts from launcher number 2 shall be taken in order and placed in trays 2 through 10 to 1; groups of parts from launcher number 3 shall be taken in order and placed in trays 3 through 10 to 2, etc. Commercial parts such as screws, nuts, washers, and pins shall be placed in the same tray as their mating or associate part. Any commercial part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. The launchers shall be reassembled using only those parts which are in the same tray.

Groups of nonmating parts

Group I

Actuator, safety (7790601)
Aperature, sight (7791011)
Arm, cocking (7791353)
Sear (779649)
Swivel, sling assembly (6147721)

Group II

Base, frame (7791012)
Extractor (7791529)
Latch (7790636)
Retainer, aperature (7791023)
Spring, helical, torsion (7791558)

Group III

Base, sight (7791013)
Lever, cocking (7790623)
Safety (7790644)
Sight, front (7791568)
Spring, helical, compression (used with firing pin) (7790656)

Group IV

Bushing, hammer spring (7790606)
Carrier, aperture (7791014)
Lock, latch (7790624)
Screw, machine, flat countersunk head (7791227)
Spring, helical, comression (used with extractor (7790655)
Stock assembly (7792767 or 11010343)

Group V

Frame assembly (7791015)
Hammer (7790622)
Screw, self-locking (7790642)
Spring, helical, compression (used with indexing plunger)(7791028)

Group VI

Barrel (7791569)
Key, windage screw (7791020)
Screw (used with stock swivel assembly)(6146873 or 11010346)
Screw, sight base (7790647)

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Group VII

Bar, safety (7790604)
Bracket, fore end (7791355)
Lock, sight (7790061)
Pin, firing (7790628)
Screw, retaining, aperature (7791024)
Spring, helical, compression used with safety actuator) 7790658)
Spring, safety (7790660)

Group VIII

Fore end (7790610)
Nut, lock, retainer (7791021)
Pivot, latch (7790638)
Screw, set, socket (7790646)
Spring, helical, torsion (7791451)

Group IX

Lug, locking, barrel (7790625)
plunger, indexing (7791022)
plunger, safety spring (7790634)
Retainer, firing pin (7790643)
Setscrew, headless, slotted (7791199)
Trigger (7790662)

Group X

Receiver (7790640) with straight headless pin (7790637)
Screw, windage (7791026)
Spring, helical, compression (used with retainer)(7791027)
Spring, sight frame (7791200)
Spring, trigger (779064l)

4.5.4.1.2 Concurrent repair parts. Concurrent repair parts shall be tested for interchangeability requirement (see 3.9.13) by disassembling two launchers previously accepted in 4.4.3.3.1.1, as necessary, and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed, and the launchers shall operate and function properly. This test may be performed independently of the launcher interchangeability test specified in 4.4.3.3.1.1 and at more frequent intervals using accepted launchers taken from current production.

4.5.4.2 Interplant. Launchers to be subjected to the interplant interchangeability test shall be given preliminary hand functioning to arrure proper operation before parts are disassembled from the launchers. Launchers shall be interchanged in a manner similar to the detailed plan in 4.5.4.1.1 except that the parts shall be divided into five groups and, that when disassembling, every other launcher used shall be one produced by a different manufacturer. The launchers shall be tested for and shall comply with the requirements for firing pin indent and firing pin intrusion before and after interchange of parts using the test methods specified in 4.5.1 and 4.5.2 respectively. In addition, the launchers shall be examined for functioning of the safety (see 3.9.1.1) and shall be tested for functioning, alinement of sights, safety actuation force, and sear safety requirements after interchange of parts using the test methods specified in 4.5.5, 4.5.6, 4.5.7} and 4.5.8 as applicable. Parts shall be identified with their manufacturer throughout the test. Before launchers are returned to the contractors, the original parts shall be reassembled to their respective launchers and the launchers given a hand functioning test to assure proper operation.

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4.5.5 Pressure resistance, functioning, and endurance tests. Launchers shall be tested for pressure resistance, functioning, and endurante using the firing fixture conforming to Drawing F7799093 assembled with parts conforming to Drawing D7799065. Test results attributable to faulty ammunition shall not be counted against the launcher being tested. The pressure resistance and functioning tests shall be performed concurrently.

4.5.5.1 Pressure resistance. Launchers shall. be tested for pressure resistance requirement (see 3.9.14) by firing one round of test ammunition (see 4.4.2). Each barrel shall have been penetrant inspected before application of final protective finish prior to this test. Launchers shall be visually examined for cracks, deformations, and other evidence of damage, and cartridge cases shall be visually examined for bulges, splits, rings, and other defects caused by defective barrels. Proof marks shall be applied as indicated on the applicable drawing on launchers that have passed this test.

4.5.5.2 Functioning. Launchers shall be tested for functioning requirement (see 3.9.15) by firing one round of test ammunition (see 4.4.4.2).

4.5.5.3 Endurance. Launchers shall be tested for endurance requirement (see 3.9.17) A total of 1,000 rounds shall be fired from the test launcher. If a malfunction or unserviceable part occurs, the contractor shall stop testing and shall take actions specified in 4.4.3.4.1 through 4.4.3.4.5.

4.5.5.3.1 Launchers shall be cleaned and oiled after each 200 rounds and at the close of each day's firing, but no part shall be altered. only parts broken or worn to the extent that they are unserviceable shall. be replaced.

4.5.5.3.2 A complete record shall be kept for each endurance test showing each malfunction and part replacement including the number of the round at which each occurred and corrective action taken.

4.5.5.8.3 At the completion of the endurance test, launchers shall be disposed of as specified in the contract (see 6.1).

4.5.6 Alinement of sight test.

4.5.6.1 Launchers shall be tested for sight alinement after the firing tests using the sight alinement fixture conforming to Drawing F8440196 in accordance with instructions contained on Drawing A8443455. With the sight assembly set at zero windage and the alinement scope mounted on the sights, the cross hairs viewed through the alinement scope shall be superimposed on the sighting image for the 200-meter position. This shall be accomplished by movement of the rear sight aperture for elevation and the front sight for azimuth. The front sight may be moved as necessary provided that it does not overhang the tenon on the barrel

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by more than .100 inch on either side; however) the front sight shall not be filed or bent. When the superimposed sight picture has been obtained vertically, the rear sight scale shall be adjusted and visually examined to determine compliance with the vertical requirement (see 3.9.16.1.) and locked in the alined meter position. The front sight shall be adjusted and visually examined to determine compliance with the horizontal requirement (see 3.9.16.1) and locked in place. The aperture shall then be positioned at the 75-meter and the 350-meter settings to determine compliance with the requirements of 3.9.16.2.

4.5.7 Safety actuation force test. Launchers shall be tested for safety actuation force requirement (see 3.9.1.2) with the launcher in a horizontal position and the safety set at the "safe" position. Inspection equipment in accordance with having B7273936 shall be used. With the force indicator gage held lightly against the comb of the stock and the V-notch end-fitting engaging the upper rear corner of the thumb seat of the safety, the load shall be applied gradually to the rear of the safety in the direction of the longitudinal axis of the launcher. The maximum load measured during the movement of the safety from the "safe" position to the "fire" position shall determine whether the safety actuation force is within the specified requirements.

4.5.8 Sear safety test. Launchers shall be tested using the indicator gage conforming to Drawing F7799102 to measure the vertical travel of sear to determine compliance with the sear safety requirement (see 3.9.1.3). The test shall be accomplished with the stock assembly removed from the launcher. During the test, the launcher shall be visually examined to assure that the trigger spring bears on the sear as specified on the applicable drawing. (When using Gage F7799102 a 0.026 movement will produce a 0.013 dial reading.

4.5.9 Packaging tests.

4.5.9.1 Determination of cleanliness. The applicable surfaces of each sample unit shall be subjected to the determination of cleanliness test in accordance with MIL-P-116. The barrel bore and chamber shall be wipe-tested for cleanliness using clean white bore cleaning swabs. The degree of cleanliness shall be verified by comparison of the test swabs with standard swab samples furinshed by the contracting officer.

4.5.9.2 Heat-sealed seam. The Level A sample unit packages shall be subjected to the heat-sealed seam test specified in MIL-P-116.

4.5.9.3 Vacuum retention.

4.5.9.3.1 A sufficient vacuum shall be drawn to cause the flexible barrier of Level A unit packages to cling snugly to the enclosed item. Care shall be exercised to insure that an excessive amount of vacuum is not applied which might cause puncture or rupture of the barrier.

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4.5.9.3.2 Interpretation of results. After remaining undisturbed at room temperature for 2 hours, the barrier shall be examined to detetmine whether it is still taut and retracts against the item when drawn away and quickly released.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack consisting of complete and packed unit and intermediate packages, as applicable, packaged in accordance with Packaging Data Sheet P8426167 to the level of protection specified in the contract (see 6.1) and packed level C shall be forwarded as specified in 3.1.

5.2 Preservation, packaging, packing, and marking. Launchers with equipment shall be preserved, unit packaged, packed, and marked in accordance-with the requirements of Packaging Data Sheet P8426167 to the level of protection specified in the contract (see 6.1). Level A unit packages shall be capable of passing the free fall drop test (see 4.3).

5.3 Repair parts. Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Lists of drawings and specifications pertinent to the launcher, showing applicable revision dates.
(c) Inspection criteria not specified herein (see 4.4.2.1.1, 4.5.2.1.2, 4.4.3.6, and 4.4.3.7.2).
(d) List of inspection equipment numbers pertinent to the launcher, showing applicable revision dates.
(e) Number of quality evaluation item samples required for tests (see 3.1).
(f) That packages opened for examination shall be repackaged by the contractor at the contractor's expense.
(g) Disposition of endurance tested launchers (see 4.5.5.3.3).
(h) Shipping instructions for quality evaluation item samples and pilot pack (see 3.1 and 5.1).
(i) Selection of applicable levels of preservation, packaging, and packing (see 5.2).
(j) Packaging instructions for repair parts (see 5.3).
(k) Place of final inspection and acceptance (see "Notes" in MIL-W-13855).
(l) List of acceptance inspection equipment to be furnished the contractor (see 4.4.4.1) and responsibilities for other Government property to be furnished the contractor.

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(m) Responsibilities for furnishing ammunition (see 4.4.4.2).
(n) Responsibility for test firing facilities and operating procedures (see 6.2).
(o) Shipping instructions for launchers when an interplant interchangeability test is required (see 4.4.3.3.2).
(p) Procedures and methods for demilitarizing and disposing of rejected material.
(q) Disposition of Government furnished property.

6.2 Test firing facilities and operating procedures should be designed by the contractor in conformance with local, state, and Federal regulations. They should be suitable fo carrying out prescribed firing tests and insure the safety of operating and visiting personnel. Copies of these contractor designs should be forwarded to the contracting officers Government facilities maybe viewed upon application to the contracting officer.

6.3 When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

6.4 To avoid delay in test firing, the Government should maintain a minimum of 2 months supply of ammunition as determined by anticipated firing requirements.

6.5 Monthly reports of the results of final examination and functioning firing, reliability and interchangeability tests should be made to Commanding General, U.S. Army Weapons Command, Attn: AMSWE-QA, Rock Islands Illinois 61.201.

6.6 Supersession data. This specification includes the requirements of Springfield Armory repurchase Description SAFD-182B dated 28 December 1964.

6.7 When warranted, the following paragraph should be included in the written contract to cover the type of quality assurance system that is desirable for this item:

Contractor's quality assurance system. The contractor shall provide and maintain a quality assurance system in accordance with MIL-Q-9858.

Custodians:
Army-WC
Navy - OS
Air Force - 84

Review activities:
Navy - OS
Air Force - 84

User activities:
Navy - MC

Preparing activity:
Army-WC

Project Number:
1010-0056

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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